

Explanation
COSC 3020
Lab Practice 11
Author: Alexander Warren
2026-04-28

For this assignment, I chose to utilize a greedy algorithm. This algorithm keeps track of local imbalances between adjacent washing machines to find the number of moves required to evenly distribute the dresses across all machines. First, we check if the number of dresses is divisible by the number of machines. If it isn't, we are unable to continue and return -1 to show it is impossible to solve the problem. Otherwise, we calculate the target value each machine should have by dividing the number of dresses by the number of machines. Finally, we loop through each machine. We keep track of the balance, which represents the number of dresses that must pass through the machine to equalize the number of dresses. The final result is how many moves it takes to reach this equilibrium.

The algorithm calculates the surplus of the current machine, how many dresses must go in or out, and lastly the cumulative imbalance, which tracks how many dresses need to flow in either direction. Whichever value is the greatest is our final result. Three test cases were used. [1,0,5], [0,3,0], and [0,2,0]. These are the example inputs shown in the assignment instructions. The program gave outputs 3, 2, and -1 respectively, which matches the expected outputs. Our time complexity is $O(n)$ because we only make one pass over the array.